Theory

Part-of-Speech (POS) Tagging is the process of assigning a grammatical category to each word in a sentence, such as **noun, verb, adjective, adverb, pronoun,** or **preposition**. It helps identify the syntactic role of words and improves the understanding of text.

Different POS taggers include **Rule-Based Taggers**, **Statistical Taggers**, **Transformation-Based Taggers**, and the **NLTK POS Tagger**, which uses a pretrained statistical model. In this experiment, the Resume Dataset is preprocessed, tokenized, and each word is assigned its corresponding POS tag using NLTK.

POS tagging is widely used in **information extraction, text summarization, machine translation, sentiment analysis, and question answering**.

In [20]:
```
import pandas as pd
import nltk
import re
from nltk.tokenize import word_tokenize
from nltk import pos_tag
nltk.download('punkt')
nltk.download('averaged_perceptron_tagger')
```

```
[nltk_data] Downloading package punkt to /home/computer/nltk_data...
[nltk_data]   Package punkt is already up-to-date!
[nltk_data] Downloading package averaged_perceptron_tagger to
[nltk_data]     /home/computer/nltk_data...
[nltk_data]   Package averaged_perceptron_tagger is already up-to-
[nltk_data]       date!
```

Out[20]:
```
True
```

In [21]:
```
df = pd.read_csv("UpdatedResumeDataSet.csv")
print(df.shape)
df.head()
```

```
(962, 2)
```

Out[21]:

	Category	Resume
0	Data Science	Skills * Programming Languages: Python (pandas...
1	Data Science	Education Details \r\nMay 2013 to May 2017 B.E...
2	Data Science	Areas of Interest Deep Learning, Control Syste...
3	Data Science	Skills â¯¢ R â¯¢ Python â¯¢ SAP HANA â¯¢ Table...
4	Data Science	Education Details \r\n MCA YMCAUST, Faridab...

In [22]:
```
print(df.columns)
df[['Category','Resume']].head()
```

```
Index(['Category', 'Resume'], dtype='object')
```

Out[22]:

	Category	Resume
0	Data Science	Skills * Programming Languages: Python (pandas...
1	Data Science	Education Details \r\nMay 2013 to May 2017 B.E...
2	Data Science	Areas of Interest Deep Learning, Control Syste...
3	Data Science	Skills â¯¢ R â¯¢ Python â¯¢ SAP HANA â¯¢ Table...
4	Data Science	Education Details \r\n MCA YMCAUST, Faridab...

In [23]:
```
sample_resume = df['Resume'].iloc[0]
```

In [24]:
```
sample_resume = sample_resume.lower()
sample_resume = re.sub(r'http\S+','',sample_resume)
sample_resume = re.sub(r'[^a-z\s]',' ',sample_resume)
sample_resume = re.sub(r'\s+',' ',sample_resume)
print(sample_resume)
```

```
skills programming languages python pandas numpy scipy scikit learn matplotlib sql java javascript jquery machine
learning regression svm na ve bayes knn random forest decision trees boosting techniques cluster analysis word emb
edding sentiment analysis natural language processing dimensionality reduction topic modelling lda nmf pca neural
nets database visualizations mysql sqlserver cassandra hbase elasticsearch d js dc js plotly kibana matplotlib ggp
lot tableau others regular expression html css angular logstash kafka python flask git docker computer vision open
cv and understanding of deep learning education details data science assurance associate data science assurance as
sociate ernst young llp skill details javascript exprience months jquery exprience months python exprience monthsc
ompany details company ernst young llp description fraud investigations and dispute services assurance technology
assisted review tar technology assisted review assists in accelerating the review process and run analytics and ge
nerate reports core member of a team helped in developing automated review platform tool from scratch for assistin
g e discovery domain this tool implements predictive coding and topic modelling by automating reviews resulting in
reduced labor costs and time spent during the lawyers review understand the end to end flow of the solution doing
research and development for classification models predictive analysis and mining of the information present in te
xt data worked on analyzing the outputs and precision monitoring for the entire tool tar assists in predictive cod
ing topic modelling from the evidence by following ey standards developed the classifier models in order to identi
fy red flags and fraud related issues tools technologies python scikit learn tfidf word vec doc vec cosine similar
ity na ve bayes lda nmf for topic modelling vader and text blob for sentiment analysis matplot lib tableau dashboa
rd for reporting multiple data science and analytic projects usa clients text analytics motor vehicle customer rev
iew data received customer feedback survey data for past one year performed sentiment positive negative neutral an
d time series analysis on customer comments across all categories created heat map of terms by survey category bas
ed on frequency of words extracted positive and negative words across all the survey categories and plotted word c
loud created customized tableau dashboards for effective reporting and visualizations chatbot developed a user fri
endly chatbot for one of our products which handle simple questions about hours of operation reservation options a
nd so on this chat bot serves entire product related questions giving overview of tool via qa platform and also gi
ve recommendation responses so that user question to build chain of relevant answer this too has intelligence to b
uild the pipeline of questions as per user requirement and asks the relevant recommended questions tools technolog
ies python natural language processing nltk spacy topic modelling sentiment analysis word embedding scikit learn j
avascript jquery sqlserver information governance organizations to make informed decisions about all of the inform
ation they store the integrated information governance portfolio synthesizes intelligence across unstructured data
sources and facilitates action to ensure organizations are best positioned to counter information risk scan data f
rom multiple sources of formats and parse different file formats extract meta data information push results for in
dexing elastic search and created customized interactive dashboards using kibana preforming rot analysis on the da
ta which give information of data which helps identify content that is either redundant outdated or trivial prefor
ming full text search analysis on elastic search with predefined methods which can tag as pii personally identifia
ble information social security numbers addresses names etc which frequently targeted during cyber attacks tools t
echnologies python flask elastic search kibana fraud analytic platform fraud analytics and investigative platform
to review all red flag cases fap is a fraud analytics and investigative platform with inbuilt case manager and sui
te of analytics for various erp systems it can be used by clients to interrogate their accounting systems for iden
tifying the anomalies which can be indicators of fraud by running advanced analytics tools technologies html javas
cript sqlserver jquery css bootstrap node js d js dc js
```

In [25]:
```
tokens = word_tokenize(sample_resume)
print(tokens[:20])
```

```
['skills', 'programming', 'languages', 'python', 'pandas', 'numpy', 'scipy', 'scikit', 'learn', 'matplotlib', 'sq
l', 'java', 'javascript', 'jquery', 'machine', 'learning', 'regression', 'svm', 'na', 've']
```

In [26]:
```
pos_tags = pos_tag(tokens)
print(pos_tags[:20])
```

```
[('skills', 'NNS'), ('programming', 'VBG'), ('languages', 'NNS'), ('python', 'VBP'), ('pandas', 'JJ'), ('numpy',
'JJ'), ('scipy', 'NN'), ('scikit', 'NN'), ('learn', 'VBP'), ('matplotlib', 'NN'), ('sql', 'NN'), ('java', 'NN'),
('javascript', 'NN'), ('jquery', 'NN'), ('machine', 'NN'), ('learning', 'VBG'), ('regression', 'NN'), ('svm', 'N
N'), ('na', 'TO'), ('ve', 'VB')]
```

In [27]:
```
for word,tag in pos_tags[:20]:
    print(word," ---> ",tag)
```

```
skills  --->  NNS
programming  --->  VBG
languages  --->  NNS
python  --->  VBP
pandas  --->  JJ
numpy  --->  JJ
scipy  --->  NN
scikit  --->  NN
learn  --->  VBP
matplotlib  --->  NN
sql  --->  NN
java  --->  NN
javascript  --->  NN
jquery  --->  NN
machine  --->  NN
learning  --->  VBG
regression  --->  NN
svm  --->  NN
na  --->  TO
ve  --->  VB
```

In [28]:
```
for word,tag in pos_tags[:20]:
    print(word," ---> ",tag)
```

```
skills  --->  NNS
programming  --->  VBG
languages  --->  NNS
python  --->  VBP
pandas  --->  JJ
numpy  --->  JJ
scipy  --->  NN
scikit  --->  NN
learn  --->  VBP
matplotlib  --->  NN
sql  --->  NN
java  --->  NN
javascript  --->  NN
jquery  --->  NN
machine  --->  NN
learning  --->  VBG
regression  --->  NN
svm  --->  NN
na  --->  TO
ve  --->  VB
```

In [29]:
```
from collections import Counter
tag_frequency = Counter(tag for word,tag in pos_tags)
print(tag_frequency)
```

```
Counter({'NN': 228, 'NNS': 91, 'JJ': 75, 'IN': 66, 'CC': 26, 'DT': 26, 'VBG': 25, 'VB': 19, 'VBP': 17, 'VBN': 13,
'TO': 11, 'VBD': 11, 'RB': 10, 'VBZ': 10, 'WDT': 7, 'MD': 3, 'FW': 2, 'CD': 2, 'PRP$': 2, 'PRP': 2, 'JJS': 1, 'PD
T': 1})
```

In [30]:
```
print("Top POS Tags\n")
for tag,count in tag_frequency.most_common():
    print(tag," :",count)
```

```
Top POS Tags

NN  : 228
NNS  : 91
JJ  : 75
IN  : 66
CC  : 26
DT  : 26
VBG  : 25
VB  : 19
VBP  : 17
VBN  : 13
TO  : 11
VBD  : 11
RB  : 10
VBZ  : 10
WDT  : 7
MD  : 3
FW  : 2
CD  : 2
PRP$  : 2
PRP  : 2
JJS  : 1
PDT  : 1
```

In [31]:
```
def perform_pos(text):
    text=text.lower()
    text=re.sub(r'[^a-z\s]',' ',text)
    tokens=word_tokenize(text)
    return pos_tag(tokens)
```

In [32]:
```
df["POS_Tags"]=df["Resume"].apply(perform_pos)
df[['Category','POS_Tags']].head()
```

Out[32]:

	Category	POS_Tags
0	Data Science	[(skills, NNS), (programming, VBG), (languages...
1	Data Science	[(education, NN), (details, NNS), (may, MD), (...
2	Data Science	[(areas, NNS), (of, IN), (interest, NN), (deep...
3	Data Science	[(skills, NNS), (r, VBP), (python, JJ), (sap, ...
4	Data Science	[(education, NN), (details, NNS), (mca, VBP), ...

In [33]:
```
df[['Category','Resume','POS_Tags']].to_csv(
    "Resume_POS_Tagged.csv",
    index=False
)
print("POS Tagged Dataset Saved Successfully.")
```

```
POS Tagged Dataset Saved Successfully.
```